

scripts/ci.sh — User Guide

Last verified: 2026-05-16 (Phase 6a + 6b §11.4.29 snake_case migration: scripts/ci.sh path-references updated from Submodules/ to submodules/) **Inheritance:** HelixConstitution §11.4.18 (script documentation mandate)

Overview

Stub doc generated from the script's in-source header comment. See scripts/ci.sh for canonical behavior. This stub exists so the CM-SCRIPT-DOCS-SYNC pre-push gate (.github/hooks/pre-push Check 9) starts gating modifications to this script.

The script's own header documentation (verbatim):

```
scripts/ci.sh - local-only CI gate for Lava.
```

Per the Local-Only CI/CD constitutional rule, this script IS the project's CI/CD apparatus. The same script runs in three modes:

<code>--changed-only</code>	Fast subset for the pre-push hook (Spotless, unit tests of changed modules, constitutional doc parser, forbidden-files check). No real-device tests; no mutation tests.
<code>--full</code>	All gates – unit tests across every module, parity gate, mutation tests where wired, fixture freshness, Compose UI Challenge Tests (requires a connected Android device or emulator). Used at tag time.
(default)	Same as <code>--full</code> .

Per Sixth Law clause 5: passing CI is necessary, NOT sufficient for a release. The operator real-device verification per Task 5.22 of SP-3a is the load-bearing acceptance gate; this script certifies the codebase is shippable, not that the user-visible feature is shipped.

Usage

See the script's in-source comment block (above) for canonical usage examples.

Pipeline steps (current order)

1. Hosted-CI forbidden-files check (Local-Only CI/CD rule)
2. Host-power forbidden-command regex check (Host Stability rule)
3. Spotless (`./gradlew spotlessCheck`)
4. Unit tests across changed modules
5. Constitutional doc parser (`scripts/check-constitution.sh`)
6. **5a1 (added 2026-05-14):** §6.AC non-fatal-coverage scan (STRICT default; closes §6.AC-debt + CM-NONFATAL-COVERAGE gate)
7. **5a2 (added 2026-05-15):** §6.AB Challenge-discrimination scan (STRICT default; closes §6.AB-debt + CM-CHALLENGE-DISCRIMINATION gate)
8. **5a3 (added 2026-05-15, 31st §6.L):** §6.AE per-feature Challenge coverage scan (STRICT default after per-feature backfill drained the queue 2026-05-15; CM-CHALLENGE-COVERAGE gate; LAVA_CHALLENGE_COVERAGE_STRICT=0 to revert to advisory)
9. **5a4 (added 2026-05-15, constitution-compliance plan Phase 1):** §11.4.32 verify-all-constitution-rules sweep (STRICT default; the §11.4.32 mandated enforcement engine for every other constitution rule; wraps every individual gate above + every hermetic test suite into one invocation; produces per-run attestation JSON at `.lava-ci-evidence/verify-all/<UTC-timestamp>.json`; CM-VERIFY-ALL-CONSTITUTION-RULES gate)
10. 5b. Hermetic bash test suites (under `tests/`)
11. (Full-mode only) Parity, mutation, fixture-freshness, Compose UI Challenge Tests on connected device

§6.AE matrix gate (separate entry point)

For §6.AE.2 gate-mode runs producing per-AVD attestation: `bash scripts/run-challenge-matrix.sh`. The runner correctly REFUSES to claim success on hosts lacking KVM (darwin/arm64) — exits 2 with a host-gap diagnostic. Real attestations require a Linux x86_64 + KVM gate-host.

Maintenance

When this script is modified, update this document in the same commit (CM-SCRIPT-DOCS-SYNC requires it). Per §11.4.18, the documentation MUST stay in sync with the codebase — no doc may be out of sync with its script.

Cross-references

- `scripts/ci.sh` — the script itself
- `docs/helix-constitution-gates.md` — gate inventory
- `HelixConstitution Constitution.md` §11.4.18 (the mandate)
- `Lava CLAUDE.md` §6.AD (HelixConstitution Inheritance)